

leakage barrier

Raw RHEED MP4
PNG frames
manual ROI

Physics features
spot / streak / connection
diffuse / stability

Frozen DINO embedding

R4 Rq estimator
automatic calibration
+ DINO residual

Predicted Rq
support / abstention

Training-group AFM
height maps

Representative AFM bank
real patch bank

Distance
2.0x predicted Rq
0.25x DINO
0.25x physics

S1 real exemplar
S4 patch synthesis
predicted-Rq scaling

Leave-one-growth-group-out
held-out group excluded
from fitting and bank

Representative morphology
Not exact local AFM reconstruction